A Paper Study of Spectral Shrunk Clustering

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Abstract

Spectral clustering builds a neighbourhood graph, embeds points with Laplacian eigenvectors, and discretizes with K-means. Chang, Nie, Ma, Yang, and Zhou (AAAI 2015) argue that a graph on the original points x_i need not regularize the embedded points f_i. Spectral Shrunk Clustering (SSC) then shrinks that embedding F on a second graph to G, drops the orthogonal constraint, and claims a convex solve that does not depend on initialization. This COMP3314 report studies the paper: the problem and contributions, the algorithm and its course connections, and the published tables and figures. A controlled reimplementation from the printed equations, without author code, does not reproduce the Table 1 SC→SSC gain at protocol γ=1.

1 Problem formulation, motivation, and contributions

High-dimensional distances are often unreliable, so many clustering pipelines first reduce dimension and then partition. Spectral clustering [Shi and Malik, 2000] finds an embedding from the spectrum of a similarity graph on the original points x_i and runs K-means there. Chang et al. observe that this two-stage pipeline assumes nearby points in the reduced space share a label, yet those neighbours may have been far apart in the original space: the manifold used to build the graph need not be the manifold of the embedding F [Chang et al., 2015].

SSC therefore keeps the usual spectral embedding F and then learns a second matrix G on a k-nearest-neighbour graph built from the rows of F. The rows g_i should stay close to f_i and should be pulled together when W_{ij} is large. The authors drop the orthogonality constraint that F inherits from the Laplacian eigenproblem, and they claim the resulting problem in G is convex and independent of initialization.

They list three merits [Chang et al., 2015]. First, the shrunk pattern has no orthogonal constraint, so it is more flexible for the low-dimensional manifold. Second, manifold learning and clustering are coupled, so the graph on F is tailored to the later K-means. Third, the algorithm is said to be convex and to converge to a global optimum, hence not to rely on initialization. Those are the paper's claims; Section 4 examines their scope.

2 Methods, algorithms, and course connections

The paper uses the symmetric normalized Laplacian of Equation (4),

$$L_n = I - D^{-1/2} A D^{-1/2}, \tag{1}$$

and obtains F from the classical normalized-cut / Ng–Shi–Malik spectral embedding [Shi and Malik, 2000, Ng et al., 2001]: the c eigenvectors for the smallest eigenvalues of L_n. A Gaussian kNN graph

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W is then built on the rows of F. After replacing a squared residual by an $\ell_{2,1}$ residual (to limit outliers; Nie et al., 2010), the printed convex problem is

$$\min_{G} \|G - F\|_{2,1} + \gamma \sum_{ij} W_{ij} \|g_i - g_j\|_2, \tag{2}$$

with $\|H\|_{2,1} = \sum_i \|h_i\|_2$ over rows. Algorithm 1 is an iteratively reweighted least-squares (IRLS) loop: form a diagonal residual weight S, reweight the graph, and solve a linear system for G. Theorem 1, as the authors state it, says that Algorithm 1 monotonically decreases (2) and, because the problem is convex, a fixed point of the printed linear update is a global minimizer of (2) [Chang et al., 2015]. Labels come from K-means on G [Lloyd, 1982] and are not written back into W or F.

COMP3314 contrasts three embeddings. PCA keeps high-variance axes. LDA uses class scatter [Ding and Li, 2007]. Spectral embedding keeps neighbourhood geometry on a graph Laplacian [von Luxburg, 2007]. SSC is a post-process of the third: it does not replace PCA or LDA, and Lloyd K-means remains the last discretization of both the spectral baseline and SSC.

3 Experimental setup and results reported in the paper

All numbers and figures in this section are *Reported in Chang et al. (2015)*, not our measurements. The paper uses eight datasets: AR (840 faces, 120 people); JAFFE (213 images, 10 subjects, 26×26); ORL (40 subjects, 10 images, 32×32); UMIST (564 images, 20 people); BinAlpha (26 letters, 30 images each); MSRA50 (1799 images, 12 classes, 32×32); YaleB (2414 images, 38 people, 32×32); and USPS (9298 digits, 16×16). Every spectral method uses $k{=}5$. Parameters of all compared methods are searched on $\{10^{-6}, 10^{-3}, 1, 10^{3}, 10^{6}\}$ and the best result is reported. The text also says each clustering is repeated 50 times with random initialization and the reported result is the one with the best objective. Tables 1–2 still print mean$\pm$standard deviation. Those two sentences are not the same protocol.

Compared methods are K-means, SC, PCA K-means, PCA-SC, LDA K-means, LDA-SC, LLC [Wu and Schölkopf, 2006], and SPLS [Hou et al., 2013]. The authors' six readings of Tables 1–2 are: (i) spectral methods beat K-means; (ii) dimensionality reduction helps; (iii) LDA beats PCA; (iv) LLC is the most robust spectral variant; (v) SPLS is second; (vi) SSC is best on every dataset and both ACC and NMI. Illustrative ACC cells include SSC 76.0 ± 1.6 vs. SC 72.7 ± 2.3 on ORL, 81.2 ± 1.6 vs. 76.1 ± 1.6 on JAFFE, 49.8 ± 1.6 vs. 34.8 ± 1.4 on YaleB, and 75.5 ± 1.9 vs. 64.3 ± 1.4 on USPS. Figure 1 plots ACC against γ and is read as preferring $\gamma{=}1$ except on MSRA50 and USPS. Figure 2, with γ fixed at 1, is read as showing a quickly decreasing objective.

4 Critical analysis

Convexity is only in G given fixed F and a nonnegative W. It does not make spectral clustering or K-means convex, and it does not certify a unique ACC or NMI. Theorem 1's global-optimum sentence is therefore a statement about the shrinkage subproblem, not about the clustering metric.

The printed stationarity condition, Equation (13), uses $\gamma\widetilde{L}$ in the linear system. Differentiating (2) when both ordered pairs of a symmetric W are stored yields $(S + 2\gamma\widetilde{L})G = SF$. Algorithm 1 line 10 replaces $\widetilde{L}$ by $\widetilde{W}$ and F by X. Those three formulae are not the same. Variant E8 is our reconstruction of (2); it is not a claim about the authors' historical binary. An existing $n{=}6$ chain check is consistent with the split: E8 matches (2) at the printed γ (max $|G|$ gap 1.17×10^{-5}); printed Equation (13) matches the $\gamma/2$ instance (max $|G|$ gap 2.33×10^{-5}). We do not execute line 10.

Several choices are unnamed. The bandwidth δ, the initializer G_0, whether undirected edges are the union or the intersection of directed kNN, and the 32×32 interpolation kernel are undefined. LDA-SC is named but not specified. Pattern shrinking is cited as SPLS [Hou et al., 2013]; we did not obtain Section 3 of that paper. "Best objective" and a table of mean$\pm$std are not one protocol.

The problem statement is clear, the objective is writable, and the method sits next to the course treatment of graph Laplacians and Lloyd K-means. Those are strengths of the paper as written. A lower ACC in our reconstruction is not, by itself, a proof that the 2015 numbers are wrong.

Table 1: Hungarian ACC and geometric NMI (%). Paper cells are reported in Chang et al. (2015), historical only. Our SC and E8 γ=1 are n=50 from `strict_reproduction_matrix.csv`. k-means++ and row-norm are not in this table.

Set (n=50)	Metric	Paper SC	Our SC	Paper SSC	Our E8 γ=1
ORL att92	ACC	72.7±2.3	56.2±4.2	76.0±1.6	57.3±3.8
	NMI	85.8±1.9	80.5±2.3	88.6±1.5	81.1±2.2
JAFFE zenodo	ACC	76.1±1.6	68.5±11.2	81.2±1.6	68.4±11.1
	NMI	80.2±0.9	83.0±5.7	84.3±1.6	83.0±5.7
YaleB cropped	ACC	34.8±1.4	18.3±1.3	49.8±1.6	17.6±1.2
	NMI	55.6±1.6	26.9±0.7	46.8±1.3	26.2±0.9
USPS libsvm	ACC	64.3±1.4	61.7±8.1	75.5±1.9	61.7±8.1
	NMI	69.5±1.6	74.9±4.5	79.8±1.6	75.2±4.6

Mean±sample SD (ddof=1) over n=50 random K-means trials (seeds $0\ldots49$, n_{init}=1) on one embedding. Paper columns are *Reported in Chang et al. (2015)*, historical only, not Our Reproduction. Protocol: Equation (4), k=5, union kNN, median directed bandwidth, raw eigenvectors, E8 γ=1, G_0=F. ORL is AT&T 92×112 bilinear 32×32 (`att92`). JAFFE is verified Zenodo (213). YaleB is CroppedYale 2414 bilinear 32×32 (`cropped2414`), not Deng Cai. USPS is LIBSVM 9298. k-means++, row-norm, LDA-SC (`APPROXIMATE`), and SPLS (`NOT_IMPLEMENTED`) are omitted. AR, UMIST-564, and MSRA50 are blocked, not filled as `EXACT`.

5 Insights and proposed extension

The one planned extension replaces union kNN W on F by mutual (AND) kNN, with the same Gaussian weights frozen before IRLS [Brito et al., 1997, Maier et al., 2009]. Convexity of (2) is unchanged; no new method and no new theorem are claimed. On JAFFE at γ=1, k=5, n=50, mutual E8 ACC is 68.4±11.2 versus union E8 68.4±11.1 (410 vs. 655 edges; 20 vs. 9 components). The outcome is negative or neutral. The useful observation is that whether shrinkage helps depends on the embedding graph and on γ, not on the slogan that a more convex subproblem must cluster better.

6 Reproduction

No author code is available. We rebuild SSC from the printed equations and tag unspecified choices as A/B/C. Datasets with a completed n=50 protocol are AT&T ORL resized bilinearly to 32×32 (`att92`) [Samaria and Harter, 1994], verified Zenodo JAFFE resized to 26×26 [Lyons et al., 1998, 2025], CroppedYale 2414 non-ambient images resized bilinearly to 32×32 (`cropped2414`) [Georghiades et al., 2001, Lee et al., 2005], and LIBSVM USPS [Hull, 1994]. Shared graph: Equation (4), k=5, union kNN, median directed bandwidth, raw eigenvectors. Primary SSC cell: E8, γ=1 frozen without labels, G_0=F, IRLS budget 300, K-means `init="random"`, n_{init}=1, seeds $0\ldots49$ on one F and one G. ACC is Hungarian matching [Munkres, 1957]; NMI is the Strehl–Ghosh geometric mean [Strehl and Ghosh, 2002]. Code: `https://github.com/SelinaMai-SYM/COMP-3314-G23.git` branch `v1`. Records: `results/strict_reproduction_matrix.csv` and `docs/final_round/`.

Table 1 quotes the paper columns as historical only and places our SC and E8 γ=1 cells beside them.

AR was not downloaded. UMIST archives yield 565 unique cropped images, not 564 [Graham and Allinson, 1998]. MSRA50 was not found. SPLS stays `NOT_IMPLEMENTED`. Those names are not filled as `EXACT`. Cropped2414 and Deng Cai are both 2414×1024 with the same class counts, but the best alignment has median L2 1.92 and no image coincides at 10^{-8}; K-means(F) ACC is 0.1834 (cropped) vs. 0.2016 (Deng Cai) against published SC 34.8%. The YaleB gap is already present in K-means on F. Paired ΔACC =ACC(G)−ACC(F) at E8 γ=1, same 50 seeds, 95% bootstrap CI reflecting only K-means noise on one fixed (F, G): ORL random +0.0109 [+0.0030, +0.0187]; JAFFE random −0.0009 (CI contains 0); YaleB random −0.0072 [−0.0097, −0.0048]; USPS random −0.0004 (CI contains 0). Only ORL's interval lies above zero, by about one percentage point; YaleB's interval lies below zero. Shrinkage does not stably help.

Chang et al. give a clear two-stage diagnosis and a writable shrinkage objective. The convexity claim applies to G given F, W, not to clustering quality, and the printed linear updates are not unique. On the four reconstructions we could run, the published SC→SSC table is not reproduced. The project is not marked fully reproduced.

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